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HPE Synergy 4-port Frame Link Module (FLM) QuickSpecs

The HPE Synergy 4-port Frame Link Module (FLM) is a 'frame manager' designed for Composable Infrastructure. The FLM efficiently manages the operations of a Synergy Frame in conjunction with HPE Composer, which is powered by HPE OneView.

HPE Synergy 4-port Frame Link Module delivers secure, scalable frame management and automated infrastructure control for composable environments.

Overview

Key functions of the HPE Synergy Frame Link Module include:

- **Scaling the environment** by linking Synergy Frames together in a management ring (single management domain),
- **Separation of the data & management networks** for management performance and resilience against Denial of Service (DoS) attacks,
- **Auto-discovery of resources** by collecting asset and inventory info of components in linked frames,
- **Control and monitoring** of the thermal and power infrastructure resources in an HPE Synergy Frame,
- **Provides out-of-band management interface to HPE OneView** to configure and manage interconnect modules.
- **Reporting the health status** for each component installed in a frame, and
- **Hosting** the HPE Synergy Console user interface.

The HPE Synergy 4-port FLM provides enhanced security in Synergy systems by implementing:

- **Secure Start** to validate the firmware to prevent tampering,
- **Hardware Root of Trust** tied to the FLM CPU for further firmware validation, and
- **File Encryption** using Trusted Platform Module (TPM) functionality.

Overview

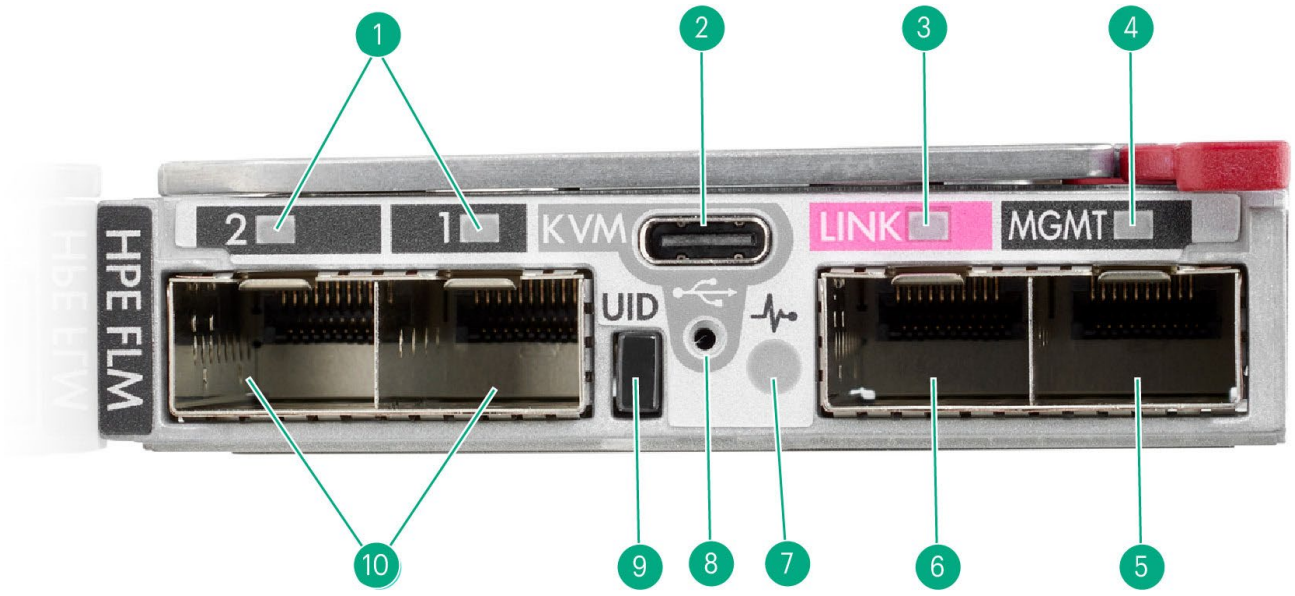


Figure 1: HPE Synergy 4-port Frame Link Module (FLM)

Item	Description	Item	Description
1.	Appliance port connectivity LEDs	6.	LINK port [SFP+][4]
2.	USB-C (KVM connection)[1]	7.	Health LED
3.	LINK port activity LED	8.	USB adapter thumb screw connection
4.	MGMT port activity LED	9.	UID LED button
5.	MGMT port [SFP+][5, 6]	10.	Appliance ports [SFP+][2, 3]

Notes:

- [1]Support for USB-C storage devices at the USB 2.0 level.
- [2]Provides a direct network connection to an appliance in bay 2. Reserved for future use.
- [3]Provides a direct network connection to an appliance in bay 1. Reserved for future use.
- [4]Use 10GbE transceivers and cables for LINK port connections.
- [5]10GbE or 1GbE based on the cable or transceiver used.
- [6]The transceiver or cable used must match the port speed of the switch used for management network connectivity.
- HPE Synergy 4-Port FLMs require supported SFP+ DAC cables or SFP+ transceivers for cabling.

Overview

What's New

The HPE Synergy Frame Link Module (FLM) is a 'frame manager' designed for Composable Infrastructure. The FLM efficiently manages the operations of a Synergy Frame in conjunction with HPE Composer, which is powered by HPE OneView.

HPE Synergy 4-port FLM is a new product version which implements enhancements for Synergy users.

Enhanced security for Synergy systems is implemented in the HPE Synergy 4-port FLM:

- **Secure Start** to validate the firmware to prevent tampering,
- **Hardware Root of Trus**t tied to the FLM CPU for further firmware validation, and
- **File Encryption** using Trusted Platform Module (TPM) functionality.

Additional capabilities in the HPE Synergy 4-port Frame Link Module (FLM) include:

- SFP+ ports to enable optical connections for reaching distant frames or remote subnets in a datacenter,
- Two additional direct-to-the-appliance bay ports which are reserved for future use, and
- USB-C connector (USB 2.0) for DisplayPort connectivity via a dongle.

Notes: HPE Synergy FLMs (both 2-port and 4-port versions) are compatible with existing applications and workloads.

Product Features

Performance

- Management (MGMT) uplink connections can operate at 10 GbE to support multiple users or administrators and to provide faster downloading of virtual media for firmware/software updates.
- Management (MGMT) port connections on the FLM are able to auto-negotiate down to 1 GbE to mate with ToR switches.
- Management rings connect multiple Synergy Frames using 10 GbE (LINK) connections.
- HPE Synergy 4-port Frame Link Modules have SFP+ ports which enable connections to distant frames and remote subnets.

Management

- HPE Synergy FLMs are 'frame managers' which operate under HPE Composer (powered by HPE OneView) management control.
- An out-of-band management interface to HPE OneView is provided by HPE Synergy FLMs to configure and manage interconnect modules.
- User Interface for the Synergy FLM includes USB-C (supported to the USB 2.0 level), Health LED, User ID (UID), and DisplayPort (via USB Adapter cable).
- Firmware for managing the HPE Synergy Frame Link Module is included in every HPE Synergy Custom SPP release, as part of HPE Synergy Software Releases.
- HPE Synergy Software Releases provide users with tested sets of firmware, drivers, and related updates to keep systems up-and-running as stable operating environments. For more information, see HPE Synergy Software Releases - Overview.

Security

All Synergy Frame Link Modules operate in FIPS (Federal Information Processing Standard) security mode. If HPE Synergy Composer (powered by HPE OneView) is put into CNSA (Commercial National Security Algorithm) mode, then HPE Composer will put the FLM into CNSA mode also.

Security enhancements which are unique to the HPE Synergy 4-Port Frame Link Module include:

- Management data stored on the frame link module is encrypted by default. The encryption is rooted in the integrated Trusted Platform Module (TPM) on the hardware device,
- Hardware Root of Trust (related to NIST SP800-193) which ensures that only firmware signed by Hewlett Packard Enterprise can be used to boot the frame link module,
- Secure Start (related to NIST SP800-193) which provides secure firmware updates,
- Firmware Resiliency prevents modification of the current running firmware image which increases reliability and by ensuring a good firmware image is always available, and
- Hardware-enforced write protection protects the frame link module from unauthorized changes.

Security features which are common to all HPE Synergy Frame Link Modules include:

- Separation of the production ('data') network from the management (or 'control') network, which protects against Denial of Service (DoS) attacks,
- Audit logging,
- Verification of certificates for Transport Layer Security (TLS),

Standard Features

- Secure protocols and certifications such as:
  - FIPS (Federal Information Processing Standard) 140-2 Level 1,
  - PCI-DSS compliance,
  - Commercial National Security Algorithm (CNSA) Suite B, and
  - General Data Protection Regulation (GDPR),
- Verified boot and Authenticated updates,
- Centralized platform security policies with Certificate management,
- No clear text passwords,
- Scope based access control (SBAC), and
- Secure out-of-the-box experience with Secure Erase of user data.

Service and Support

For the most up-to-date information on HPE Services, please refer to the HPE Services – Supplemental QuickSpecs, which provides a comprehensive and regularly updated overview of available services.

Configuration Information

Models

Notes:

- **HIGHLY RECOMMENDED:** Add a second HPE Synergy FLM into each Synergy Frame for redundancy.
- **CAUTION:** Both FLM units of a redundant pair in a Synergy Frame must be of the same generation.

Description	SKU
HPE Synergy 4-port Frame Link Module	876852-B21
Notes: This Synergy Frame Link Module has 4 ports (LINK port, MGMT port, two Appliance ports).	
HPE Synergy 4-port Frame Link Module USB Adapter	P06458-B21
Notes: USB Adapter ('dongle') is required for DisplayPort access to the rear of the HPE Synergy Frame (USB-C to DisplayPort).	

Compatibility

HPE Synergy 4-port Frame Link Module supports the following hardware.

HPE Synergy Frames

HPE Synergy 12000 Frame

HPE Synergy Interconnects

- HPE Virtual Connect SE 100GB F32 Module for HPE Synergy
- HPE Virtual Connect SE 40GB F8 Module for HPE Synergy
- HPE Synergy 40Gb F8 Switch Module

HPE Cables and Transceivers

- See the 'Related Options' section of this document.

Notes:

- **CAUTION:** Solution integrity when using distant system connections must be verified – including limiting communication latencies in the system to <5 milliseconds (one way) in the solution.
- **CAUTION:** Cable type and transceivers determine the distance limit of connections.
  - DAC cable lengths @ 10Gbps = 15 meters (max)
  - AOC cable lengths using SR transceivers = 300m (max)
  - Fibre cable lengths using SR transceivers = 300m (max)
  - Fibre cable lengths using LR transceivers = 10km (max)
  - Fibre cable lengths using ER transceivers = 40km (max)

Supported Configurations



Figure 2: HPE Synergy 12000 Frame - Rear View
(Frame Link Modules shown partially extended)

Item	Description
1.	Frame Link Module #1
2.	Frame Link Module #2

HPE Synergy Frame Link Modules install into one of two frame link module bays in the rear of the frame.

Notes:

- **HIGHLY RECOMMENDED:** Add a second Synergy FLM be added into each Synergy Frame for redundancy.
- **CAUTION:** Both FLM units of a redundant pair in a Synergy Frame must be of the same generation.
- **CAUTION**: Use caution when installing a Synergy FLM into the Synergy Frame to avoid damage to the connector. Installing a Synergy FLM with a damaged connector can result in damage to the Synergy Frame midplane.

Related Options

DisplayPort USB Adapter

DisplayPort connection to rear of HPE Synergy 4-port Frame Link Module

Description	SKU
HPE Synergy 4-port Frame Link Module USB Adapter	P06458-B21

Notes: USB Adapter ('dongle') is required for DisplayPort access to the rear of the HPE Synergy Frame (USB-C to DisplayPort).

Copper Cables (DAC)

DAC Cables connect SFP+ ports using copper cable with built-in transceivers.

Notes:
- For use between Synergy 4-port FLMs, and for connecting management uplinks to other SFP+ network connections.
- Direct Attach Cables (DAC) use copper cables with built-in transceivers on each end.
- **CAUTION:** Cable type determines the distance limit of connections.
- Active DAC cables @ 10Gbps = 15 meters (max).
- Passive DAC cables = ~5 meters.

Description	SKU
HPE BladeSystem c-Class 10GbE SFP+ to SFP+ 3m Direct Attach Copper Cable	487655-B21
HPE BladeSystem c-Class 10GbE SFP+ to SFP+ 5m Direct Attach Copper Cable	537963-B21
HPE Networking X240 10G SFP+ SFP+ 0.65m DAC Cable	JD095C
HPE Networking X240 10G SFP+ SFP+ 1.2m DAC Cable	JD096C
HPE Networking X240 10G SFP+ SFP+ 3m DAC Cable	JD097C
HPE Networking X240 10G SFP+ SFP+ 5m DAC Cable	JG081C
HPE Networking X240 10G SFP+ 7m DAC Cable	JC784C

Active Optical Cables (AOC)

AOC Cables connect SFP+ ports using fibre cables with built-in transceivers.

Notes:
- For use connecting between Synergy 4-port FLMs and for connecting management uplinks to other SFP+ network connections.
- HPE Synergy 4-port FLMs have SFP+ 10Gbps 2-fibre ports.
- **CAUTION**: AOC cable lengths using SR transceivers = 300m (max).

Related Options

Transceivers

Transceiver use should consider bandwidth, port types, and port counts.

Notes:
- HPE Synergy 4-port FLMs have SFP+ 10Gbps 2-fibre ports.
- **CAUTION:** Solution integrity when using distant system connections must be verified – including limiting communication latencies in the system to <5 milliseconds (one way) in the solution.
- **CAUTION:** Cable type and transceivers determine the distance limit of connections.
  - Extended distances can be implemented using fibre cables with HPE SR, LR, and ER transceivers.
  - Fibre cable lengths using SR transceivers = 300m (max)
  - Fibre cable lengths using LR transceivers = 10km (max)
  - Fibre cable lengths using ER transceivers = 40km (max)

Transceivers (10 Gbps) for connecting SFP+ ports using AOC fibre cables.

Description	SKU
HPE BladeSystem c-Class 10Gb SFP+ SR Transceiver	455883-B21
Notes: Typical transceiver for use with single-mode fibre cable (short range).	
HPE Networking X130 10G SFP+ LC SR Transceiver	JD092B
HPE BladeSystem c-Class 10Gb SFP+ LR Transceiver	455886-B21
Notes: Typical transceiver for use with single-mode fibre cable (long range).	
HPE Networking X130 10G SFP+ LC LR Transceiver	JD094B
HPE Networking X130 10G SFP+ LC ER 40km Transceiver	JG234A

Transceivers (10 Gbps) for connecting SFP+ ports to RJ45 ports using CAT6A cables.

Description	SKU
HPE 10GBase-T SFP+ Transceiver	813874-B21

Notes:
- Typical 10GbE transceiver for connecting CAT6A cables to SFP+ ports on the HPE Synergy 4-port FLM.

Transceivers (1 Gbps) for connecting SFP+ ports (for use with MGMT ports only).

Description	SKU
HPE BladeSystem c-Class Virtual Connect 1G SFP SX Transceiver	453151-B21
Notes: Typical 1Gb transceiver for use with fibre cable (<100m).	
HPE BladeSystem CClass Virtual Connect 1G SFP RJ45 Transceiver	453154-B21
Notes: Typical 1Gb transceiver for connecting CAT cables to SFP+ ports on the HPE Synergy 4-port FLM.	

Related Options

Fibre Cables

Fibre Cables require transceivers to connect SFP+ ports.

Notes:

- HPE Synergy 4-port FLMs have SFP+ 10Gbps 2-fibre ports.
- **CAUTION:** Solution integrity when using distant system connections must be verified – including limiting communication latencies in the system to <5 milliseconds (one way) in the solution.
- **CAUTION:** Cable type and transceivers determine the distance limit of connections.
  - Extended distances can be implemented using fibre cables with HPE SR, LR, and ER transceivers.
  - Fibre cable lengths using SR transceivers = 300m (max)
  - Fibre cable lengths using LR transceivers = 10km (max)
  - Fibre cable lengths using ER transceivers = 40km (max)

OM3 Fibre Cables

Description	SKU
HPE LC to LC Multi-mode OM3 2-Fiber 0.5m 1-Pack Fiber Optic Cable	AJ833A
HPE LC to LC Multi-mode OM3 2-Fiber 1.0m 1-Pack Fiber Optic Cable	AJ834A
HPE LC to LC Multi-mode OM3 2-Fiber 2.0m 1-Pack Fiber Optic Cable	AJ835A
HPE LC to LC Multi-mode OM3 2-Fiber 5.0m 1-Pack Fiber Optic Cable	AJ836A
HPE LC to LC Multi-mode OM3 2-Fiber 15.0m 1-Pack Fiber Optic Cable	AJ837A
HPE LC to LC Multi-mode OM3 2-Fiber 30.0m 1-Pack Fiber Optic Cable	AJ838A
HPE LC to LC Multi-mode OM3 2-Fiber 50.0m 1-Pack Fiber Optic Cable	AJ839A

OM4 Fibre Cables

Description	SKU
HPE Premier Flex LC/LC Multi-mode OM4 2 Fiber 1m Cable	QK732A
HPE Premier Flex LC/LC Multi-mode OM4 2 Fiber 2m Cable	QK733A
HPE Premier Flex LC/LC Multi-mode OM4 2 Fiber 5m Cable	QK734A
HPE Premier Flex LC/LC Multi-mode OM4 2 Fiber 15m Cable	QK735A
HPE Premier Flex LC/LC Multi-mode OM4 2 Fiber 30m Cable	QK736A
HPE Premier Flex LC/LC Multi-mode OM4 2 Fiber 50m Cable	QK737A

HPE Synergy Services

Deployment/Installation & Start-up Services

Description	SKU
HPE Factory Express Synergy Initial Frame Package 4 Service	HA454A1-300
HPE Factory Express Synergy Add-on Frame Package 4 Service	HA454A1-301
HPE Synergy First Frame Onsite Startup Service	U8JM3E
HPE Synergy Addl Frame Onsite Startup Service	U8JM4E

Technical Specifications

HPE Synergy Frame Link Module			
Shipping Dimensions	Length	18.50 in (469.9 mm)	
	Width	10.13 in (257.3 mm)	
	Height	2.00 in (50.8 mm)	
Shipping Weight		2.13 lbs (0.97 kg)	
Product Hardware Specifications	**Indicators**	Health LED	
		UID LED	
		MGMT & LINK Port Activity LEDs	
		Appliance Port Connectivity LEDs	
	Environmental Ranges	**Temperature Range**	
		Operating	10°C to 35°C (50° to 95°F)
		Non-Operating	-30°C to 60°C (-22°F to 140°F)
		Relative Humidity (non-condensing)	
		Operating	10% to 90% relative humidity (Rh), 28°C (82.4°F) maximum wet bulb temperature, non-condensing.
		Shipping	5% to 95% relative humidity (Rh), 38.7°C (101.7°F) maximum wet bulb temperature, non-condensing.
		Notes: – Operating temperature has an altitude derating of 1.8°F (1°C) per 1,000 ft (304.8 m). No direct sunlight allowed. Upper operating limit is 10,000 ft (3,048 m) or 70 KPa/10.1 psia. Upper non-operating limit is 30,000 ft (9,144 m) or 30.3 KPa/4.4 psia. – Storage maximum humidity of 95% is based on a maximum temperature of 113°F (45°C). Altitude maximum for storage is 70 KPa.	
Power Specification	12V @ 4.2A (50 W max)		

Environmentally friendly Products and Approach

End-of-life Management and Recycling

Hewlett Packard Enterprise offers end-of-life Hewlett Packard Enterprise product return, trade-in, and recycling programs in many geographic areas. For trade-in information, please go to http://www.hpe.com/info/recycle**.** To recycle your product, please go to: http://www.hpe.com/info/recycle or contact your nearest Hewlett Packard Enterprise sales office. Products returned to Hewlett Packard Enterprise will be recycled, recovered or disposed of in a responsible manner.

The EU WEEE directive (2002/95/EC) requires manufacturers to provide treatment information for each product type for use by treatment facilities. This information (product disassembly instructions) is posted on the Hewlett Packard Enterprise web site at: http://www.hpe.com/info/recycle. These instructions may be used by recyclers and other WEEE treatment facilities as well as Hewlett Packard Enterprise OEM customers who integrate and re-sell Hewlett Packard Enterprise equipment.

Summary of Changes

Date	Version History	Action	Description of Change
21-Sep-2026	Version 9	Changed	QuickSpecs maintenance updates were completed, and the Related Options section was updated.
20-Jul-2026	Version 8	Changed	Service and Support section was updated.
16-Feb-2026	Version 7	Changed	HPE Rebranding applied.
19-Feb-2024	Version 6	Changed	Networking product names were updated.
04-Dec-2023	Version 5	Changed	Service and Support section was updated.
15-Nov-2021	Version 4	Changed	Service and Support Pointnext Tech Care and Complete Care information updated.
16-Sep-2019	Version 3	Changed	QuickSpecs was updated.
16-Sep-2019	Version 2	Changed	Configuration Information, Related Options, and Technical Specifications sections were updated.
		Added	SKUs added in Configuration Information and Related Options.
03-Sept-2019	Version 1	Created	New QuickSpecs.

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